My research aims to advance and integrate artificial intelligence (AI), computational optimization, and high-performance and GPU computing to enable the future of transportation, logistics, and critical infrastructure through three interconnected research pillars:

1. AI-integrated discrete optimization in transportation and logistics
2. Sequential decisions in critical infrastructure under uncertainty
3. Extreme-scale optimization on next-generation computing systems

This statement outlines my contributions within each pillar and my current and future work on methodological and applied problems across engineering, management, and public policy. It concludes with a discussion of my existing collaborations and potential funding opportunities.

1 AI-integrated discrete optimization in transportation and logistics

Trucks carry more than 12.8 billion tons of U.S. freight a year [1] over networks planned at multiple time scales, from the strategic siting of facilities that last for decades, through the tactical sizing of fleets for months, to the operational routing and assignment of orders within the day. Yet decisions at these scales interact. For example, the operational routes and their cost are a function of where facilities are sited and how fleets are sized. Ignoring this interaction can produce plans that appear economical but prove costly to execute. Depots are sited assuming each customer is served from the nearest one, fleets are sized with routing costs from static formulas, and orders are assigned to drivers assuming they follow the prescribed routes [2].

A key difficulty in modeling these multiscale interactions is that the induced operational cost can be *(i)* computationally expensive to evaluate (as in vehicle routing or scheduling), *(ii)* available only through black-box calls to a simulator (such as an agent-based model), or *(iii)* observable only through prior evaluations (such as the route choices of drivers in historical data). Embedding such costs within a higher-level planning model is thus challenging, as they often lack a mathematical representation, and even when they have one, the exact formulations do not scale to real-world transportation networks [3].

My goal is to integrate supervised machine learning with mixed-integer optimization [4][1]. I develop sparse neural surrogates of operational costs, embed them exactly within higher-level planning models, and use classical algorithms to optimize planning decisions (Figure 1). My contributions include [2, 6–8]:

1. **Neural embedded optimization for location-routing at scale.** I designed a specialized neural architecture that preserves the permutation invariance property of routing costs, and embedded the trained network exactly [5] within the mixed-integer optimization model for location-allocation decisions [6, 7], which led to the Best Paper Award of the IISE Logistics and Supply Chain Division. Experiments on four benchmark sets provide empirical evidence that my method is *(i)* accurate (within several percent of the best known solutions), *(ii)* fast (**more than an order of magnitude faster than existing heuristics** on large instances), and *(iii)* modular (the same trained network handles any number of customers and depots). I am extending this line of work to a *family of neural embedded algorithms* for multi-depot, two-echelon and periodic routing, the problems behind municipal waste collection, school bus routing and mobile healthcare [9].

[1] This integration is often called constraint learning or predict-then-optimize [5], and is well suited to combinatorial settings.

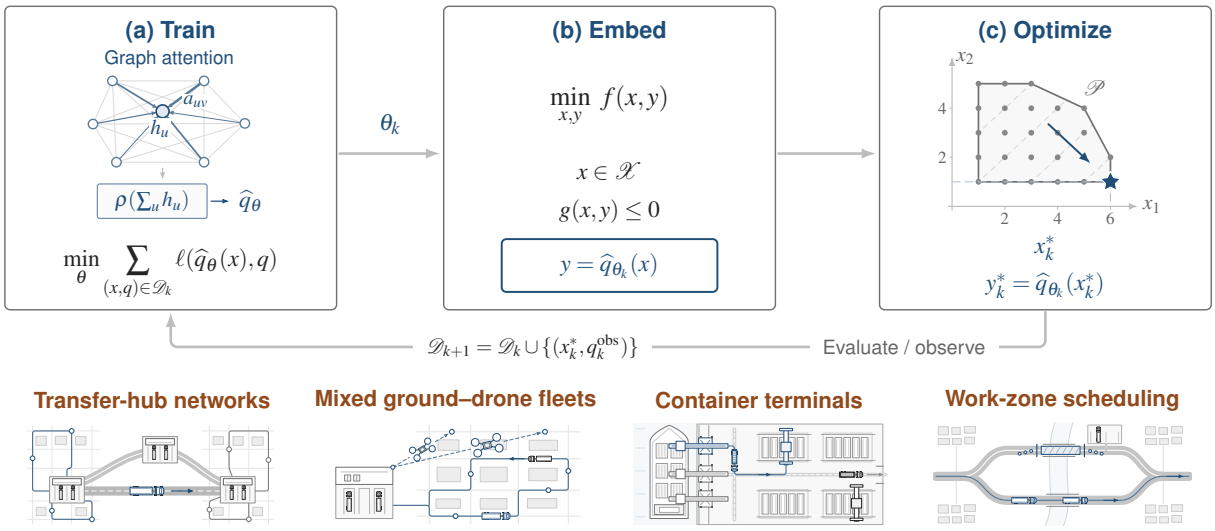


Figure 1: Vision for the first research pillar on AI-integrated discrete optimization.

2. **Unified neural estimators of routing costs.** I developed a novel multi-set representation of routing instances that allows a sum decomposition of the optimal cost across customers, vehicles and depots. Using this representation, I designed the **first unified neural estimator** of routing cost for fast what-if analyses in logistics, such as sizing a school bus fleet, designing service zones or setting vehicle size limits [8]. Experiments on five routing variants (with capacities, time windows, heterogeneous fleets and multiple depots) show state-of-the-art accuracy in estimating routing costs, with errors two- to fifteen-fold smaller than those of the best baselines [8].
3. **Data-driven on-time last-mile delivery.** I developed a method for assigning orders to drivers in last-mile delivery, where drivers often deviate from planned routes [2]. From historical deliveries, the method learns the total time drivers took to deliver a batch of orders, and embeds this estimate within an order assignment mixed-integer optimization model. Experiments on two months of food delivery data from Shanghai show that the method **reduces late deliveries threefold relative to a routing-based baseline** [2].

Methodological extensions. I plan to extend this work in three directions. First, I will retrain or fine-tune the surrogate with the gap between the cost of the decisions it induces and the true optimal cost [4]. Existing methods differentiate through a linear program. However, in my discrete setting this means repeated solves of a mixed-integer model at each training step, which is computationally intractable and an open research question. Second, the most common embedding of the neural surrogate introduces one binary variable per ReLU neuron [5]. However, this can make the planning model computationally intractable for deep networks, and I aim to continue my ongoing research on penalty alternating direction methods, which need no additional binary variables [10]. Further, I also want to investigate the embedding of trained diffusion and language models. Third, I will establish provable lower and upper bounds on the optimal cost of the original planning problem from the solution of the neural embedded model [7].

Engineering applications. I plan to apply my neural embedded location-allocation models [6, 7] to optimize hub locations and customer assignments in **transfer-hub networks** (Figure 1).

Autonomous trucks would carry freight between hubs on long-haul routes and delivery vans would serve the last mile. I will train neural networks to estimate local routing costs from each hub's location and assigned customers, then embed these networks in the design model to minimize the total cost of hub investment and transportation. For **mixed ground–drone fleets**, I intend to build on my heterogeneous drone routing work [11] to train neural networks that estimate delivery operating costs from customer orders and fleet composition. I will embed these networks in a fleet design model that determines fleet size and composition to minimize total acquisition and operating costs.

I also see opportunities to extend this methodology to **container-terminal capacity** and **work-zone scheduling**. For container terminals, I would build on my queue-based infrastructure planning work [3] by learning from terminal simulations how yard crane allocations affect mean ship turnaround times and embedding these approximations in capacity investment models with turnaround-time constraints. For work zones, I would learn from traffic simulations how road-closure schedules affect traffic delay and incorporate these estimates in maintenance scheduling models.

2 Sequential decisions in critical infrastructure under uncertainty

In this pillar, I aim to optimize the planning and operation of critical transportation infrastructure, including EV charging networks, hydrogen refueling corridors and rail classification yards. The decisions must hedge against uncertainty and adapt as new information becomes available. Relevant uncertainties include network-level link flows and induced demand, vehicle arrivals and charging or refueling times at stations, and train arrivals and processing times in rail yards.

Most existing approaches either make myopic decisions (neglecting future consequences) or plan against a point forecast of the underlying uncertainty. This can lead to unreliable operations, with long queues and congestion at stations, delays at depots and missed connections in rail yards. Further, the uncertainty may evolve in response to decisions (*decision-dependent*). For example, building charging stations in stages along a highway corridor can induce additional EV flow.

The unified framework of Powell [12] provides a principled approach to sequential decision-making under uncertainty (Figure 2). I aim to design mixed-integer optimization policies that scale to real-world transportation networks and minimize the cost of the current decision along with a tractable approximation of future cost from the resulting state. These policies will provide feasibility guarantees under capacity and service constraints. For interpretability, I will examine constraint slacks and use sensitivity analysis to assess how capacity limits affect current decisions and the ability to respond to uncertain future demand.

My past and ongoing work [3, 13, 14] with Argonne National Laboratory and Mitsubishi Electric Research Laboratories motivates this pillar. Below, I summarize my contributions to infrastructure planning and fleet operations and outline my ongoing work and future research directions.

1. **At the strategic level** [3], I designed models for locating charging stations and deciding how many chargers of each type to install, subject to limits on the expected time vehicles spend waiting and charging. I modeled each station and charger type as a multiserver queue with stochastic arrivals and service times. I used Lagrangian dual decomposition on high-performance computers to obtain feasible plans for the Chicago metropolitan region. In comparisons of infrastructure and access policies, I show that combining existing depots with newly built stations for shared use by fleet operators **nearly halves the cost of building and**

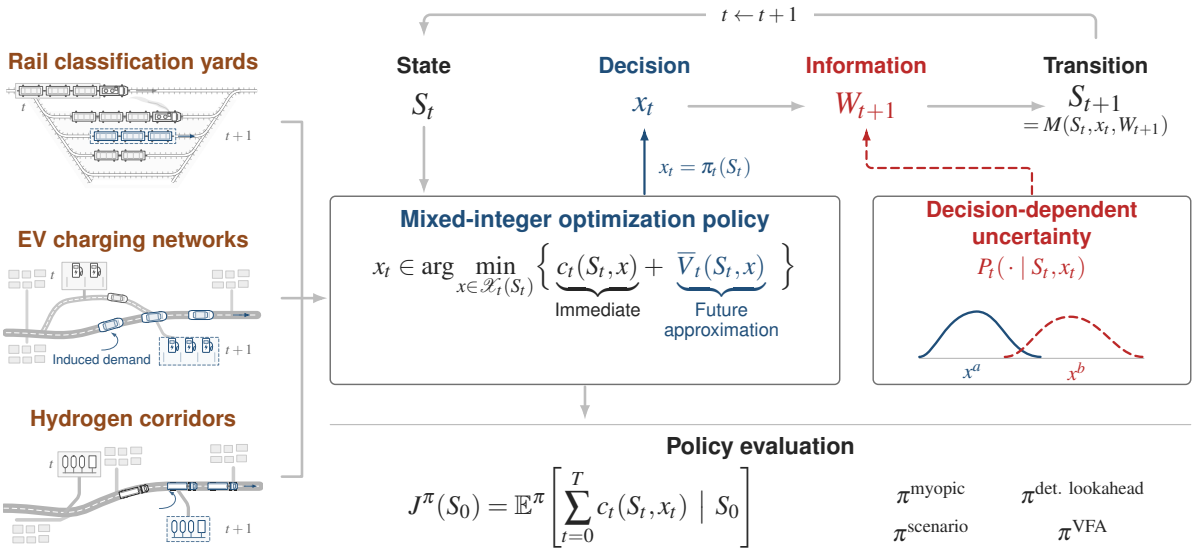


Figure 2: Vision for the second research pillar on sequential decisions in critical infrastructure under uncertainty.

using the charging network relative to plans that restrict each operator to the newly built stations it owns. In ongoing work, we are extending these models to plan charging networks for long-distance EV travel, where station and charger investments influence demand. The resulting distributionally robust model admits an exact mixed-integer linear reformulation for a decision-dependent ℓ_1-Wasserstein ambiguity set.

2. **At the tactical level** [13], I designed a sequential freight market-clearing policy for a truckload spot market in which a neutral broker matches loads to trucks in multi-load schedules under federal hours-of-service regulations, as load requests and bids become known over time. The mixed-integer optimization policy minimizes immediate cost together with an approximation of future value, with learned marginal truck values that depend on their positions and availability. I developed and implemented a specialized branch-price-and-cut algorithm to solve each weekly market clearing exactly. In a case study on a market synthesized from Argonne's POLARIS freight model, my approach **increases the freight market surplus by two-thirds over myopic market clearing**.
3. **At the operational level** [14], I developed a mixed-integer linear programming model for depot charging of heterogeneous fleets that couples vehicle assignments with charging and storage dynamics under power and jockey movement constraints. I also developed a tractable piecewise linear inner approximation of charging-power limits as a function of battery state of charge. This work was conducted during my appointment in the optimization group at Mitsubishi Electric Research Laboratories. The resulting schedules maximize the number of vehicles charged by their deadlines, then minimize delays, and finally minimize electricity cost. Mitsubishi Electric has **filed a joint patent application** with the U.S. Patent and Trademark Office. We are extending the model to sequential charging decisions under uncertain arrival times and states of charge.

Methodological extensions. I plan to extend my work on freight market clearing and infrastructure planning in two directions. First, my freight market-clearing policy [13] approximates the cost of

serving future loads (see Figure 2) by adding individual truck values, although trucks may compete for the same future loads. I will approximate this cost for the fleet as a whole with neural models (see Pillar 1) and extend branch-price-and-cut to retain separate scheduling subproblems for each truck. Second, I will extend our static EV charging-network planning model [15] to sequential decision-making, developing neural decision rules for capacity expansion under decision-dependent uncertainty in demand.

Engineering applications. I intend to extend my EV charging-network planning models [3] to the phased expansion of EV charging networks and hydrogen refueling corridors. These models will determine when, where and how much capacity to install as demand responds to earlier investments, subject to limits on expected waiting times. In rail classification yards, I will adapt my scheduling approaches [13, 14] to assign tracks and sequence car movements as freight trains arrive at the yard. These policies will account for uncertain arrivals and processing times and the capacity needed by later trains, with the aim of reducing missed outbound connections.

3 Extreme-scale optimization on next-generation computing systems

In this pillar, my past and ongoing work includes extreme-scale network design for last-mile delivery [3], GPU-native primal heuristics, and GPU-accelerated very large-scale neighborhood search for railroad blocking. For brevity, I do not elaborate on these projects here. My future work includes the following directions.

Network design extensions. I will extend my current work on extreme-scale network design by developing hardware-aware decomposition methods that coordinate computations across CPUs, GPUs and AI accelerators. These methods will balance workloads subject to memory and communication limits, with the aim of obtaining better network designs within a fixed computing budget.

New algorithmic directions. I plan to pursue two further directions within this research pillar. First, I will develop algorithms to solve neural embedded mixed-integer models directly on GPUs and AI accelerators (see Pillar 1). Keeping the trained networks and optimization computations on the same device can reduce data transfers and support the repeated solves needed to retrain the neural surrogates. Second, I seek to develop mixed-precision algorithms for general mixed-integer programming, motivated by the growing performance advantage of low-precision matrix arithmetic on AI accelerators.

Research collaborations and funding opportunities

My doctoral research has involved collaborations with researchers across national laboratories, academia, and industry. My collaborators at Argonne National Laboratory include Dr. Jeffrey Larson (Mathematics and Computer Science Division), Dr. Omer Verbas, Dr. Abdelrahman Ismael, Dr. Olcay Sahin and Dr. Hyunseop Uhm. I also collaborate with Prof. Taner Cokyasar (Texas A&M University and Argonne National Laboratory) and Prof. Tanveer Hossain Bhuiyan (University of Texas at San Antonio). My doctoral research on neural embedded optimization, advised by Prof. Anirudh Subramanyam at Penn State, is in collaboration with Prof. Changhyun Kwon and Doyoung Lee (KAIST and Omelet, Inc.) and Dr. Henri Lefebvre (LIRMM, CNRS). At Mitsubishi Electric Research Laboratories, I worked on charging schedules with my mentor, Dr. Arvind U. Raghunathan, and am extending this work with Dr. Anna Deza. My work on GPU-accelerated optimization is with Prof. Aleksandr Kazachkov (University of Florida).

The U.S. Department of Transportation projects that the value of U.S. freight will double between

2020 and 2050 [16]. The White House's AI Action Plan calls for wider AI adoption in energy and investment in AI and robotics for manufacturing and logistics [17]. Furthermore, the U.S. Marine Corps identifies logistics distribution and advanced logistics analytics as priorities in its Force Design Update [18]. These priorities highlight federal interest in applying AI to transportation and logistics, which aligns with my research on AI-integrated optimization.

I plan to pursue funding through NSF's Advanced Manufacturing, Design, and Operations (AMDO) program for AI-integrated optimization in manufacturing and supply chains, and its Engineering for the Built Environment (EBE) program for infrastructure decisions under uncertainty. For scalable optimization algorithms, I see opportunities through NSF's Computational Mathematics program and the DOE Office of Science's Advanced Scientific Computing Research (ASCR) program. To develop reusable optimization software, I will consider future opportunities through NSF's Cyberinfrastructure Systems for Scientific Innovation (CSSI) program.

Finally, I will seek support from DOT's University Transportation Centers, state departments of transportation, industry and nonprofit organizations, and pursue research subcontracts from national laboratories.

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